

# **Lecture 03b – One-way ANOVA**

ENVX2001 Applied Statistical Methods

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**Why ANOVA?**

## The problem with multiple $t$ -tests

Suppose we want to compare weight gain of chicks fed on **4 different diets**.

With  $t$ -tests, we'd need  $\binom{4}{2} = 6$  pairwise comparisons:

1-2, 1-3, 1-4, 2-3, 2-4, 3-4

Even if there are no true differences, each test has a 5% chance of incorrectly finding significance. How does that add up?

```
prob_all_correct ← 0.95^6  
prob_all_correct
```

```
[1] 0.7350919
```

We have a **26.5%** chance that at least one test result will be incorrect!

## We need a better method

The **problem of multiple comparisons** requires a technique that considers **all** treatments at once.

## ANOVA – Analysis of Variance

Before deciding *which* treatments differ, we first consider:

1. Differences **between** the treatment groups – **between-treatment effects**
2. Differences **within** each treatment group – **within-treatment effects**
  - due to random environmental fluctuations, genetics, experimental error

## Chick weight data

## The experiment



- 20 chicks
- 4 diets
- 5 replicates per diet
- Response: weight gain (g)

```
chicks_wide ← read.csv("data/chicks.csv")  
chicks_wide
```

|   | Diet.1 | Diet.2 | Diet.3 | Diet.4 |
|---|--------|--------|--------|--------|
| 1 | 99     | 61     | 42     | 169    |
| 2 | 88     | 112    | 97     | 137    |
| 3 | 76     | 30     | 81     | 169    |
| 4 | 38     | 89     | 95     | 85     |
| 5 | 94     | 63     | 92     | 154    |

## Reshaping the data

```
chicks <- read.csv("data/chicks.csv") |>
  pivot_longer(
    cols = starts_with("Diet"),
    names_to = "diet",
    values_to = "weight"
  ) |>
  mutate(diet = as.factor(diet))
```

chicks

```
# A tibble: 20 × 2
  diet    weight
  <fct>   <int>
1 Diet.1     99
2 Diet.2     61
3 Diet.3     42
4 Diet.4    169
5 Diet.1     88
```

|    |        |     |
|----|--------|-----|
| 6  | Diet.2 | 112 |
| 7  | Diet.3 | 97  |
| 8  | Diet.4 | 137 |
| 9  | Diet.1 | 76  |
| 10 | Diet.2 | 30  |
| 11 | Diet.3 | 81  |
| 12 | Diet.4 | 169 |
| 13 | Diet.1 | 38  |
| 14 | Diet.2 | 89  |
| 15 | Diet.3 | 95  |
| 16 | Diet.4 | 85  |
| 17 | Diet.1 | 94  |
| 18 | Diet.2 | 63  |
| 19 | Diet.3 | 92  |
| 20 | Diet.4 | 154 |



# **ANOVA concepts**

## Terminology

- **Factor** (or treatment): the categorical variable of interest (here: diet)
- **Levels**: the categories within a factor (here: Diet 1, Diet 2, Diet 3, Diet 4)
- **Replicates**: the number of observations per level (here:  $r = 5$ )
- $t$  = number of treatments;  $N$  = total observations =  $r \times t = 20$

This is a **one-way** (or one-factor) ANOVA because there is only one factor (diet).

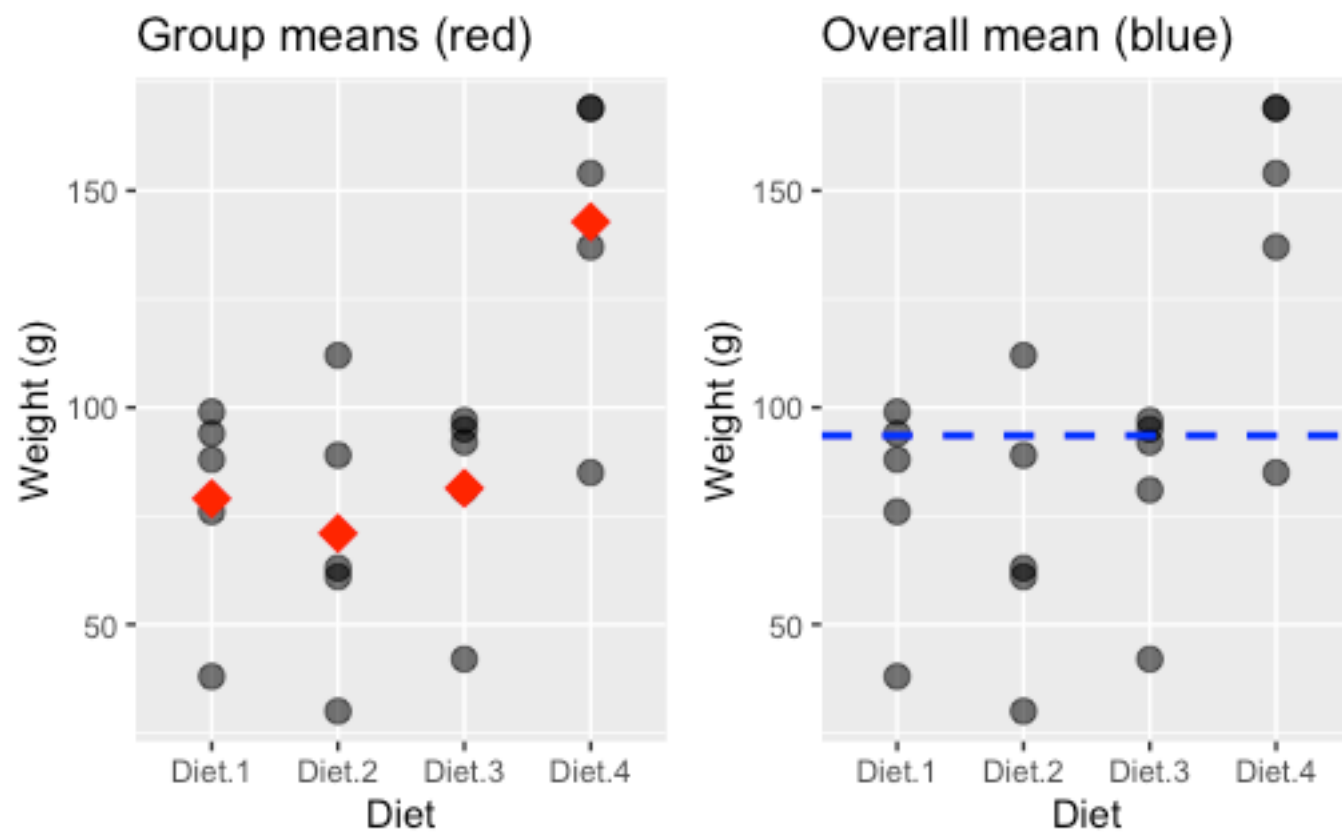
## Which model best describes the data?

```
overall_mean ← mean(chicks$weight)
group_means ← chicks |>
  group_by(diet) |>
  summarise(mean_wt = mean(weight))

p1 ← ggplot(chicks, aes(diet, weight)) +
  geom_point(size = 3, alpha = 0.6) +
  geom_point(data = group_means, aes(diet, mean_wt),
    shape = 18, size = 5, colour = "red") +
  labs(title = "Group means (red)", x = "Diet", y = "Weight (g)")

p2 ← ggplot(chicks, aes(diet, weight)) +
  geom_point(size = 3, alpha = 0.6) +
  geom_hline(yintercept = overall_mean, colour = "blue",
    linewidth = 1, linetype = "dashed") +
  labs(title = "Overall mean (blue)", x = "Diet", y = "Weight (g)")

library(patchwork)
p1 + p2
```



Does the **group means model** (left) explain the data better than the **overall mean model** (right)?

## Model equation

## Same form as the $t$ -test

$$y_{ij} = \mu_i + \varepsilon_{ij}$$

- $i = 1, 2, \dots, t$  (treatment);  $j = 1, 2, \dots, n_i$  (replicate)

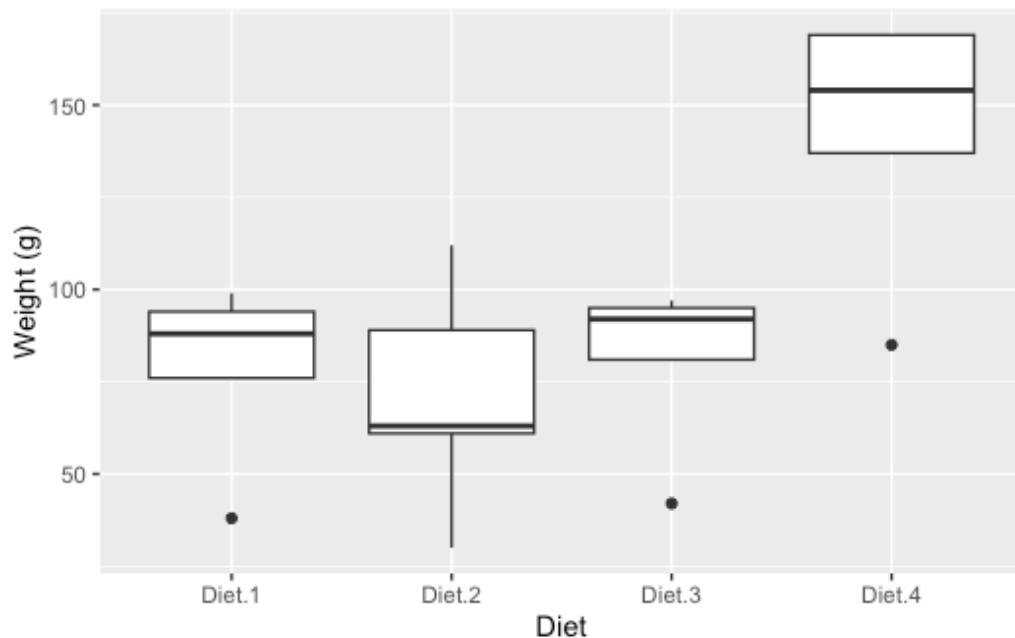
In the chick example:

- $y_{ij}$  = observed weight gain for the  $j$ th chick on Diet  $i$
- $\mu_i$  = mean weight gain for chicks on Diet  $i$
- $\varepsilon_{ij}$  = random error (residual)

# Assumptions

## Checking normality

```
ggplot(chicks, aes(diet, weight)) +  
  geom_boxplot() +  
  labs(x = "Diet", y = "Weight (g)")
```



```
shapiro.test(chicks$weight)
```

Shapiro-Wilk normality test

data: chicks\$weight  
W = 0.93272, p-value = 0.1742

$p > 0.05$ : we can assume normality.



## Checking equal variances

- $\sigma_1^2 = \sigma_2^2 = \dots = \sigma_t^2$
- General guide:  $\frac{\text{largest SD}}{\text{smallest SD}} < 2.0$
- Alternatively, use **Bartlett's test** of homogeneity of variance:

```
bartlett.test(weight ~ diet, data = chicks)
```

Bartlett test of homogeneity of variances

data: weight by diet

Bartlett's K-squared = 0.85164, df = 3, p-value = 0.8371

### Warning

Bartlett's test is **sensitive to non-normality**. Only use it when normality is reasonable.

# Hypothesis test

## Hypotheses

- **Null hypothesis:**  $H_0 : \mu_1 = \mu_2 = \dots = \mu_t$
- **Alternative hypothesis:**  $H_1$ : not all  $\mu_i$  are equal

! Important

ANOVA only tells us that **at least two** group means are different – not *which* ones.

## The concept: partition variability

Partition the total variability into components:

$$SS_{\text{total}} = SS_{\text{treatment}} + SS_{\text{residual}}$$

- $SS_{\text{treatment}}$ : variation **between** groups (due to diet)
- $SS_{\text{residual}}$ : variation **within** groups (random error)

If the treatment has a real effect,  $SS_{\text{treatment}}$  should be **large** relative to  $SS_{\text{residual}}$ .

# Variance partitioning

**Total sum of squares:**  $SS_{\text{total}}$

$$SS_{\text{total}} = \sum (y_{ij} - \bar{y})^2$$

```
overall_mean ← mean(chicks$weight)
ss_total ← sum((chicks$weight - overall_mean)^2)
ss_total
```

```
[1] 29678.95
```

## Treatment sum of squares: $SS_{\text{treatment}}$

$$SS_{\text{treatment}} = \sum n_i \times (\bar{y}_i - \bar{y})^2$$

```
group_means <- chicks |>
  group_by(diet) |>
  summarise(grp_mean = mean(weight))

ss_trt <- sum(5 * (group_means$grp_mean - overall_mean)^2)
ss_trt
```

```
[1] 16466.95
```

## Residual sum of squares: $SS_{\text{residual}}$

$$SS_{\text{residual}} = \sum (y_{ij} - \bar{y}_i)^2$$

```
chicks_with_means <- chicks |>
  left_join(group_means, by = "diet")

ss_res <- sum((chicks_with_means$weight - chicks_with_means$grp_mean)^2)
ss_res
```

```
[1] 13212
```

**Check:**  $SS_{\text{total}} = SS_{\text{treatment}} + SS_{\text{residual}}$

```
c(ss_total, ss_trt + ss_res)
```

```
[1] 29678.95 29678.95
```



## **ANOVA table**

## Structure

| Source       | df      | SS                  | MS                        | F                                 |
|--------------|---------|---------------------|---------------------------|-----------------------------------|
| Treatment    | $t - 1$ | $SS_{\text{trt}}$   | $SS_{\text{trt}}/(t - 1)$ | $MS_{\text{trt}}/MS_{\text{res}}$ |
| Residual     | $N - t$ | $SS_{\text{res}}$   | $SS_{\text{res}}/(N - t)$ |                                   |
| <b>Total</b> | $N - 1$ | $SS_{\text{total}}$ |                           |                                   |

- $N$  = total observations,  $t$  = number of treatment levels
- **Mean Squares** (MS) standardise SS so they are comparable
- The larger the ratio  $MS_{\text{trt}}/MS_{\text{res}}$ , the stronger the treatment effect

## Using `aov()` in R

```
model ← aov(weight ~ diet, data = chicks)
summary(model)
```

```
          Df Sum Sq Mean Sq F value Pr(>F)
diet         3  16467    5489   6.647  0.004 **
Residuals    16  13212     826
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

## Interpreting the results

- $F = 6.65$  with  $df = 3, 16$
- $p = 0.004$  which is  $< 0.05$ , so we **reject**  $H_0$
- There are **significant differences** in mean weight gain amongst the 4 diets

## How much variability is explained?

$$\frac{SS_{\text{treatment}}}{SS_{\text{total}}} = \frac{1.6467 \times 10^4}{2.9679 \times 10^4} = 55.5\%$$

The diets explain about 55% of the total variability in chick weight gain.

**Post-hoc: which groups differ?**

## Confidence intervals for group means

We can examine 95% CIs for each treatment mean:

$$95\% \text{ CI} = \bar{y}_i \pm t_{N-t}^{0.025} \times SE(\bar{y}_i)$$

where  $SE(\bar{y}_i) = \sqrt{MS_{\text{res}}/n_i}$

## Using emmeans

```
library(emmeans)
emm ← emmeans(model, "diet")
emm
```

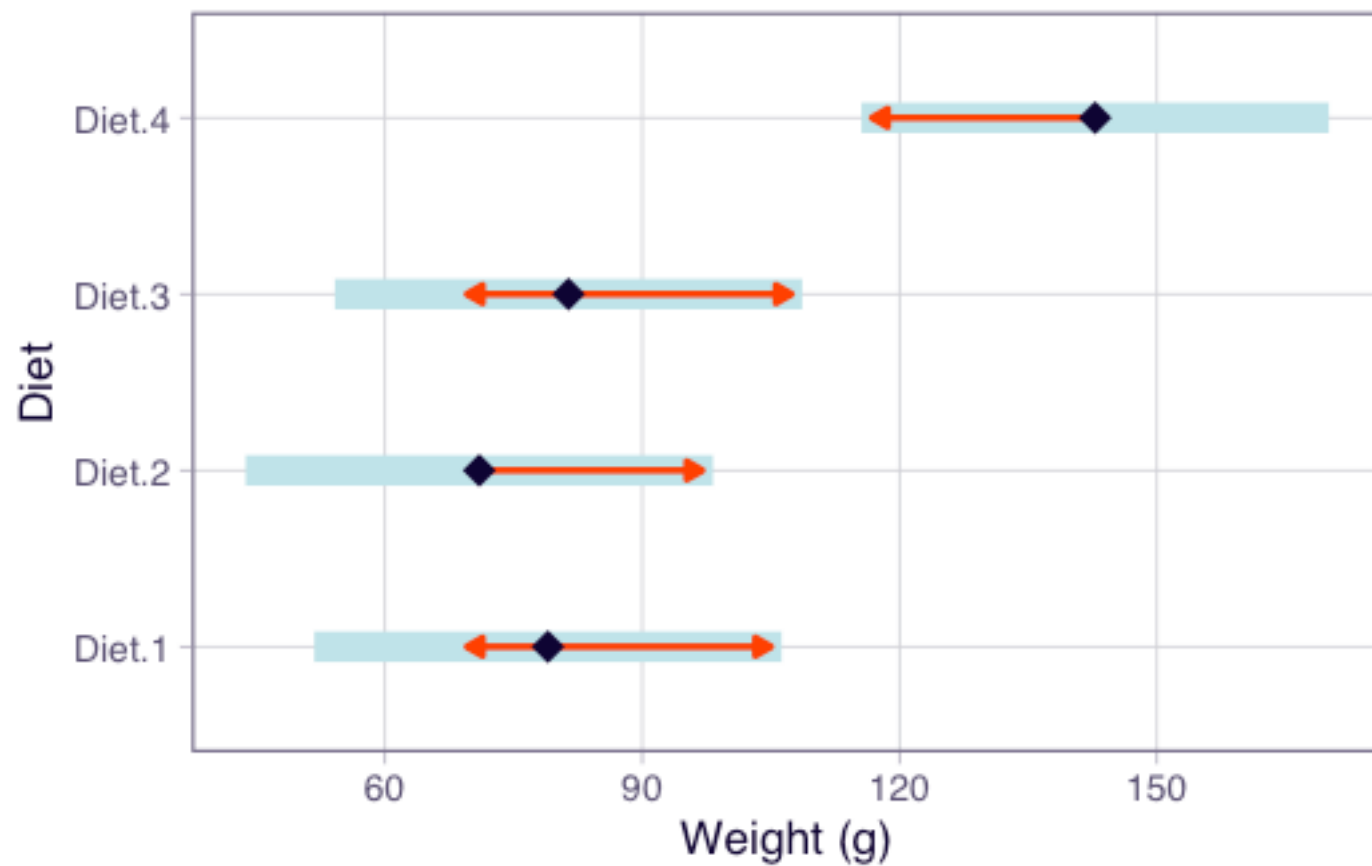
| diet   | emmean | SE   | df | lower.CL | upper.CL |
|--------|--------|------|----|----------|----------|
| Diet.1 | 79.0   | 12.9 | 16 | 51.8     | 106.2    |
| Diet.2 | 71.0   | 12.9 | 16 | 43.8     | 98.2     |
| Diet.3 | 81.4   | 12.9 | 16 | 54.2     | 108.6    |
| Diet.4 | 142.8  | 12.9 | 16 | 115.6    | 170.0    |

Confidence level used: 0.95

## Visualising group comparisons

```
plot(emm, comparisons = TRUE) +  
  labs(x = "Weight (g)", y = "Diet")
```





Groups with **overlapping** arrows are **not** significantly different from each other.

# Summary

## $t$ -test vs ANOVA

- $t$ -test: compares means of **2** groups
- **ANOVA**: compares means of **2 or more** groups simultaneously
- ANOVA avoids the inflated Type I error rate from multiple  $t$ -tests
- The ANOVA  $F$ -test tells us *if* differences exist, but not *where*
- Post-hoc methods (e.g. `emmeans`) identify *which* pairs differ

## Next week

- How to better identify which pairs are significantly different
- How to test and interpret model assumptions using **residual diagnostics**

## References

- Quinn & Keough (2002), Chapter 7: Section 7.1
- Mead et al. (2002), Chapter 18: Sections 18.1–18.3

**Thanks!**

**Questions?**

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